GROSS DOMESTIC PRODUCT

How India Calculates GDP

Concepts, Estimation Methods, and Where India Stands
July 2026

Source: MOSPI — National Accounts Statistics: Sources & Methods; Quarterly GDP Compilation Methodology

What Is GDP, and Why Does It Matter?

Definition

GDP measures the total value of production activity in an economy. It is a production concept: the sum of gross value added by all resident producer units, plus taxes less subsidies on products not already included in valuation of output.

Why it matters

- Determines how much a country can afford to consume
- Directly affects the level of employment
- Consumption of goods & services — individual and collective — is central to community welfare

A key nuance: Residency, not geography

GDP is not production within a country's geographic boundary — it is production by resident producer units. A resident firm's team working temporarily abroad contributes to that firm's exports, not to the GDP of the country where the work happens.

Example

An Indian engineering firm sends staff abroad to install equipment. That output counts as India's export — not as the GDP of the country where the installation occurs.

Source: MOSPI, National Accounts Statistics – Sources & Methods 2007, Ch. 8, ¶8.1–8.2

Three Equivalent Approaches to Measuring GDP

Production Approach

Sum of Gross Value Added (Output – Intermediate Consumption) across all economic activities, plus indirect taxes less subsidies on products.

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GVA at basic prices +
Net taxes on products
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Income Approach

Sum of the incomes generated by production: compensation of employees, operating surplus/mixed income, plus taxes net of subsidies on production.

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Compensation + Surplus +
Net production taxes
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Expenditure Approach

Final use (demand) of output: government & private consumption, capital formation, changes in stock, and net exports.

Private Final Consumption (PFCE)
Government Final Consumption (GFCE)
Gross Fixed Capital Formation (GFCF)

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GFCE + PFCE + GFCF +
ΔStocks + Net Exports
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India treats the Production approach as the firmer estimate; the discrepancy with Expenditure-side GDP is presented explicitly in the National Accounts.

Source: MOSPI, National Accounts Statistics – Sources & Methods 2007, Ch. 8, ¶8.2

Production Approach: How Sectors Are Actually Estimated

QGVA = Output − Intermediate Consumption, per industry — using the benchmark-indicator method (extrapolating prior-year GVA with a physical/volume indicator)

Sector	Key Indicator / Data Source	Estimation Logic
Crops	Quarterly crop production (Min. of Agriculture)	Extrapolate prior-year output with production growth
Livestock	Milk/egg/meat/wool targets (Dept. of Animal Husbandry)	Growth in production applied to prior-year GVA
Mining & Quarrying	Coal/crude/gas output, IIP-Mining, listed co. financials	Volume extrapolation + corporate sector via staff cost/PBT
Manufacturing	Index of Industrial Production (IIP), BSE/NSE financials	Organised sector via listed-co. growth; unorganised via IIP
Construction	Cement production, steel consumption, IIP	Extrapolated using cement/steel/glass volume growth
Trade & Repair	Sales tax data, corporate financials	Quarterly sales-tax growth + private corporate growth
Financial Services	RBI bank deposits & credit, insurance premia	Deposit/credit growth deflated by GVA deflator
Public Administration	Govt. revenue expenditure (net of interest)	Extrapolated using deflated revenue expenditure growth

Source: MOSPI, Methodology of Compiling Quarterly GDP Estimates (2017), Table: Quarterly Estimates of GVA by Production Approach

Direct vs. Indirect Estimation: Filling the Data Gaps

Not every component of GDP can be measured directly. Roughly 10% of GDP estimates rely on indirect 'rates and ratios' — proxy relationships used where direct price or production data doesn't exist.

Direct Estimation

Used when both production/output data AND prices are directly observed — value of output = production × price.

Example: Principal crops (~49 crops) — value of output is prepared directly from DESMOA production data and State DES market prices during peak marketing season.

Indirect Estimation

Used when direct data is missing — a proxy 'rate or ratio' relative to a known/similar item stands in for the missing figure.

- Small millets valued at 75% of the weighted average price of jowar, bajra, barley, maize & ragi
- Other pulses valued at 85% of the weighted average price of arhar/urad/moong/masoor/horsegram
- Processed tea = 22.5% of raw tea-leaf output; bagasse = 22.5% of gur production
- Fuel-wood output (chronically under-reported) is estimated from NSS consumption-expenditure data instead of forest department production records

Source: MOSPI, 13.3 Gross Domestic Product — 'Rates and Ratios used in the estimation of GDP by industry', ¶13.3.61-13.3.71

Known Data Gaps & Limitations (MOSPI's Own Assessment)

Agriculture	Yield rates for minor crops, grass & fodder are based on very old norms (some from 1952-53); quarterly estimates still follow a crop-calendar rather than actual production.
Livestock	No annual data on animal deaths; population estimates rely on infrequent Livestock Censuses not synchronised across states, causing drift after droughts or shocks.
Forestry	Fuel-wood production is 'totally unreliable' from Forest Department records due to unauthorised extraction — estimated instead from household consumption surveys.
Unorganised sector (broad)	Manufacturing, trade, and services in the informal economy depend on benchmark Enterprise Surveys / Economic Census follow-ups conducted only once every 5 years; interim years use physical indicators (IIP, workforce growth) as proxies.
Prices	Agricultural prices are collected at peak marketing-period rates, not first-point-of-transaction prices — introducing a systematic approximation into farm-gate value estimates.

Source: MOSPI, 13.3 Gross Domestic Product — 'Data Gaps in the estimates of GDP by industry', ¶13.3.38-13.3.60

Real GDP vs. Nominal GDP: What's the Difference?

Nominal GDP

GDP valued at current-year (prevailing) prices. Captures growth in both the quantity of goods/services produced AND changes in their prices (inflation).

Nominal growth = Real growth + Inflation (GDP deflator)

FY 2025-26 (Provisional Estimates): Real GDP grew 7.6% while Nominal GDP grew 8.6% — the ~1 percentage-point gap reflects economy-wide price change (the GDP deflator).

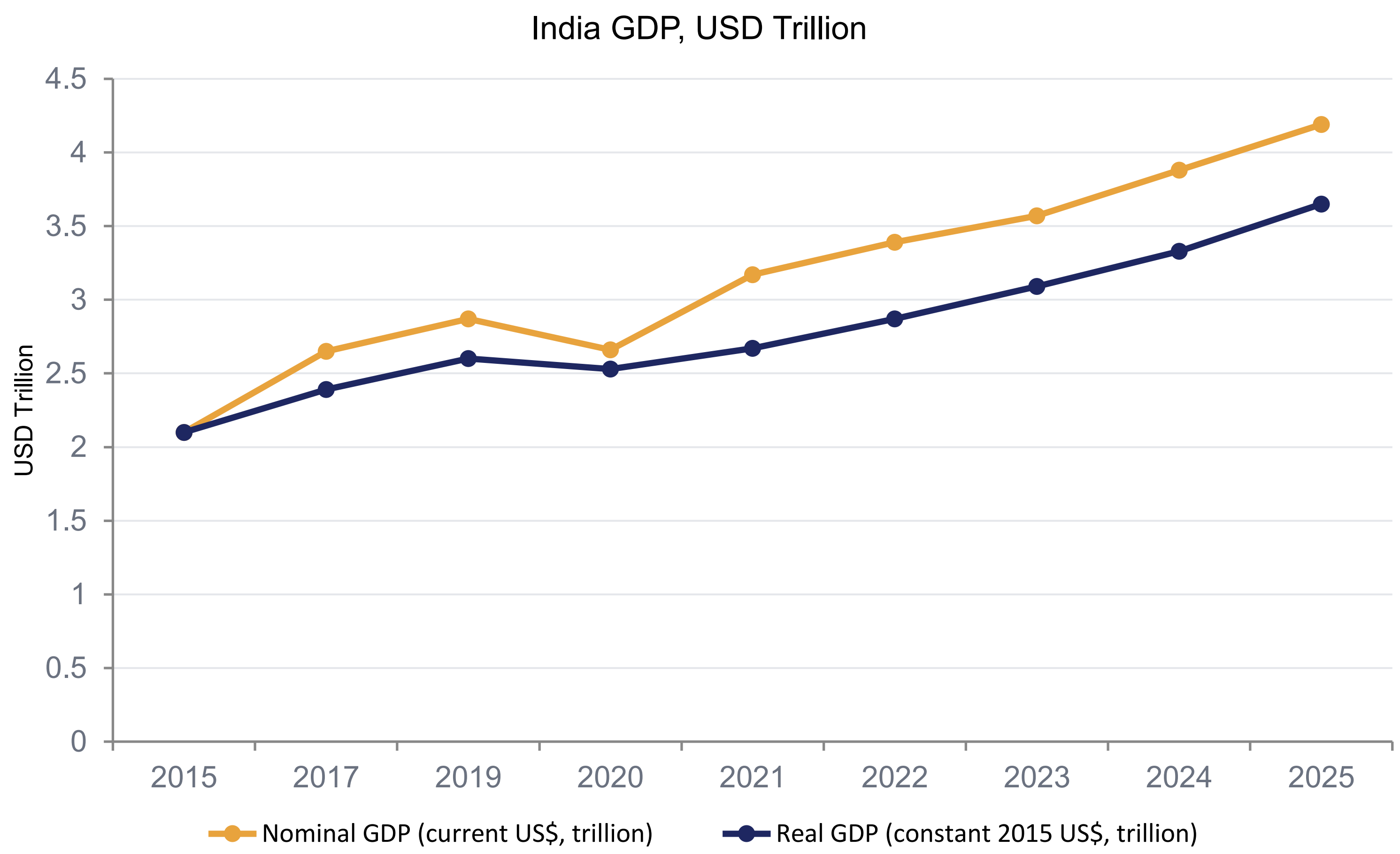
Real GDP

GDP valued at the prices of a fixed base year. Strips out the effect of price changes so only genuine volume/output growth remains — the figure used for 'growth rate' headlines.

India's current base year: 2022-23 (previously 2011-12)

Source: MOSPI, Press Note on New Series of GDP Estimates with Base Year 2022-23 (Feb 2026); Provisional Estimates FY2025-26 (Jun 2026)

India's GDP: Nominal vs. Real (2015–2025)



Reading the gap

The widening gap between the two lines is cumulative inflation. Nominal GDP roughly doubled between 2015 and 2025, but real (volume) growth was more moderate once price effects are removed.

India's nominal GDP crossed $4 trillion in 2025, making it the world's 4th-5th largest economy by nominal terms.

Illustrative trend based on World Bank (World Development Indicators), IMF WEO, and MOSPI data. Figures approximate.

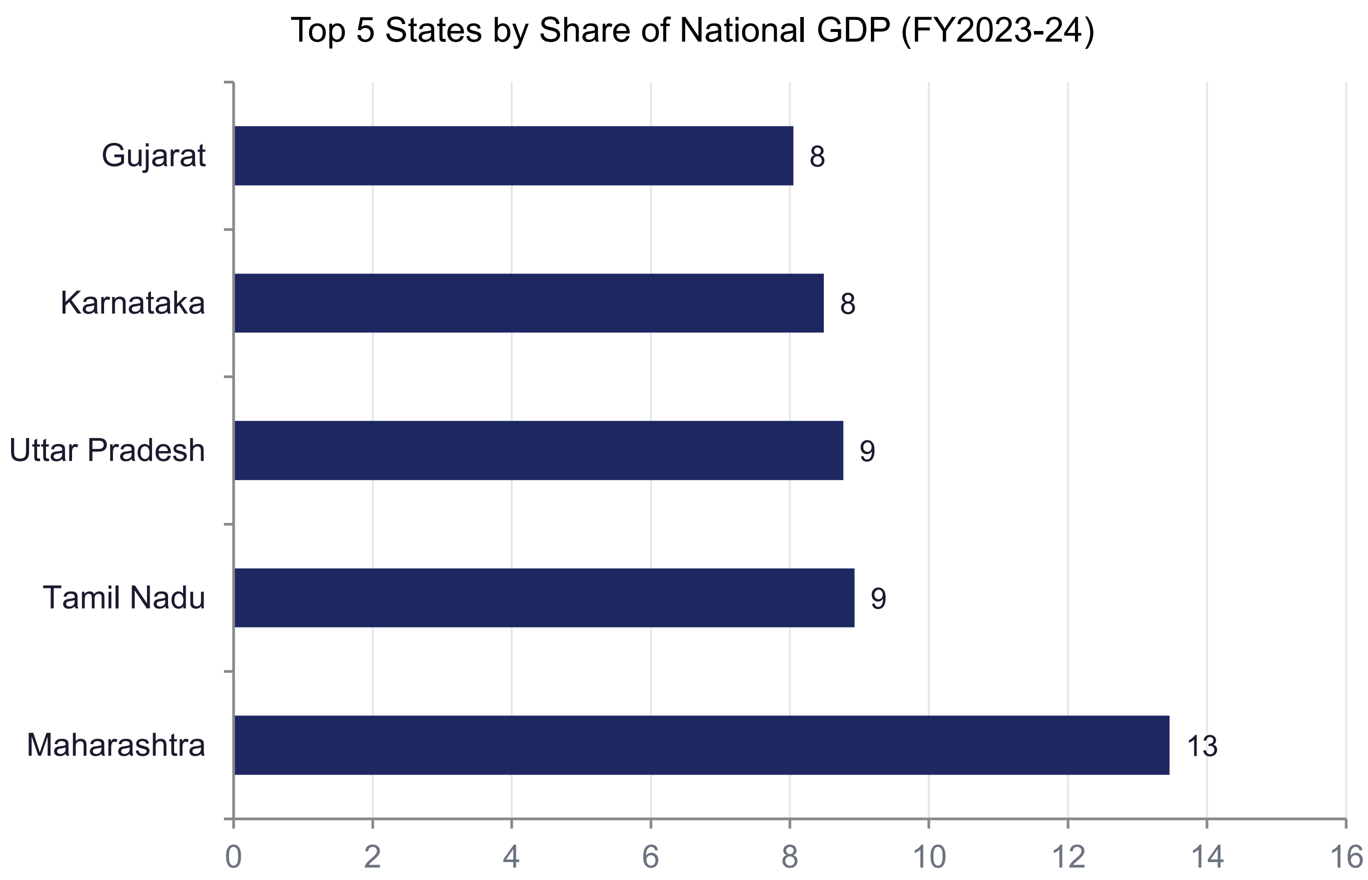
India in Global Context: GDP & GDP Per Capita, 2025

Country	Nominal GDP (US$ trillion)	Population (approx.)	GDP per Capita (US$, approx.)
United States	30.62	335 million	~91,400
China	19.4	1.41 billion	~13,750
Germany	5.01	84 million	~59,600
Japan	4.28	124 million	~34,500
India	**4.19**	**1.45 billion**	**~2,900**
United Kingdom	3.96	69 million	~57,400

India is the world's 4th–5th largest economy by nominal GDP, but ranks far lower on a per-capita basis — reflecting its large population base. On a PPP basis, India is the 3rd-largest economy globally.

Sources: Statista/IMF WEO (Oct 2025 & Apr 2026), Worldometer/IMF nominal GDP rankings 2025. Population and per-capita figures approximate.

GSDP: How India's Economic Output Is Distributed Across States



Key facts

- GSDP = the summation of value of all goods & services produced within a state during a year — computed by NSO/state Directorates of Economics & Statistics using the same National Accounts framework
- Top 5 states together account for ~47.7% of India's GDP
- Southern states (Karnataka, AP, Telangana, Kerala, Tamil Nadu) collectively contribute ~30.6% of GDP
- Sikkim & Goa lead on GDP per capita; Bihar has among the lowest

Source: StatisticsTimes.com (MOSPI/NSO state GSDP data), FY2023-24 nominal GDP shares

Beyond Official Statistics: Estimating Sub-Regional GDP from Nightlights

Source: Prakash, A., Shukla, A.K., Bhowmick, C., & Beyer, R.C.M. (2019), RBI Occasional Papers, Vol. 40

Night-time Luminosity: Does It Brighten Understanding of Economic Activity in India?

Prakash, Shukla, Bhowmick & Beyer — RBI Occasional Papers (2019)

- Night-light data show reasonably robust correlation with GDP, industrial production, and credit growth at the national level
- Quarter-on-quarter growth in night lights tracks GDP growth reasonably well
- Night lights are strongly correlated with Gross State Domestic Product (GSDP)
- The 'inverse Henderson elasticity' of night lights w.r.t. GSDP is statistically significant, though smaller than global estimates
- Useful where formal-sector data lags or misses informal economic activity — data are near-real-time, low-cost, and spatially granular

Summary

1. **Theory:** GDP = value of production by resident units; measured via Production, Income, or Expenditure — three equivalent lenses.
2. **Estimation:** Each GVA and expenditure component is estimated using distinct real-world indicators — IIP, sales tax, BoP data, and more.
3. **Aggregation:** Production-side GVA + net taxes on products = GDP; deflation via WPI/CPI converts constant-price estimates to current prices.
4. **Context:** India's real GDP grew 7.6% in FY2025-26; nominal GDP crossed ~$4.1 trillion — the world's 4th-5th largest economy, though per-capita income remains modest.
5. **Granularity:** GSDP and nightlights-based estimation extend the picture below the national level, into states and sub-regions.